

⚠ LEG 3 • AUTOMOTIVE AI

THE AI ROAD TRIP

LEG 3

8 more self-driving-car AI systems. Built in 8 days.

Swipe to take Exit 01 →



01

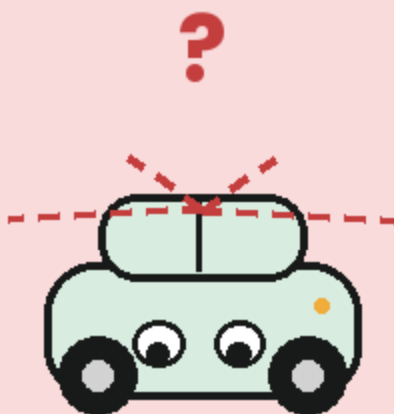
EXIT 01 • SENSOR FUSION

Knows which instinct to trust, moment to moment

Four classical physics models and one neural network each guess where the car goes next. This AI learns, timestep by timestep, which guess to lean on — physics in a straight line, the learned model through a tricky turn.

Like a driver who trusts muscle memory on a straight road, but leans on instinct through a sharp bend.

OLD



NEW

**Better trajectory guesses in hard turns**

EKF-Neural-Fusion

02

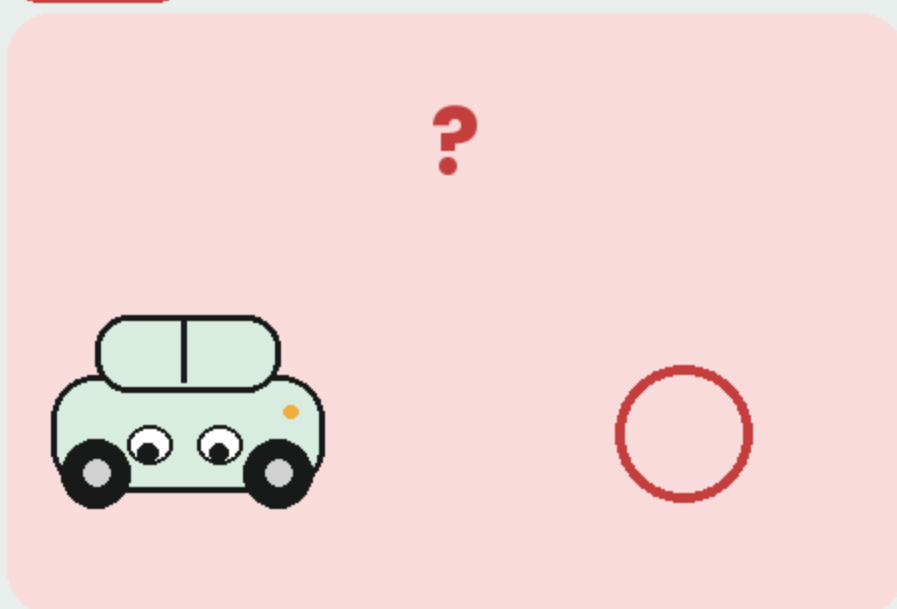
EXIT 02 • RISK-AWARE PLANNING

Replays how the road ahead will actually unfold

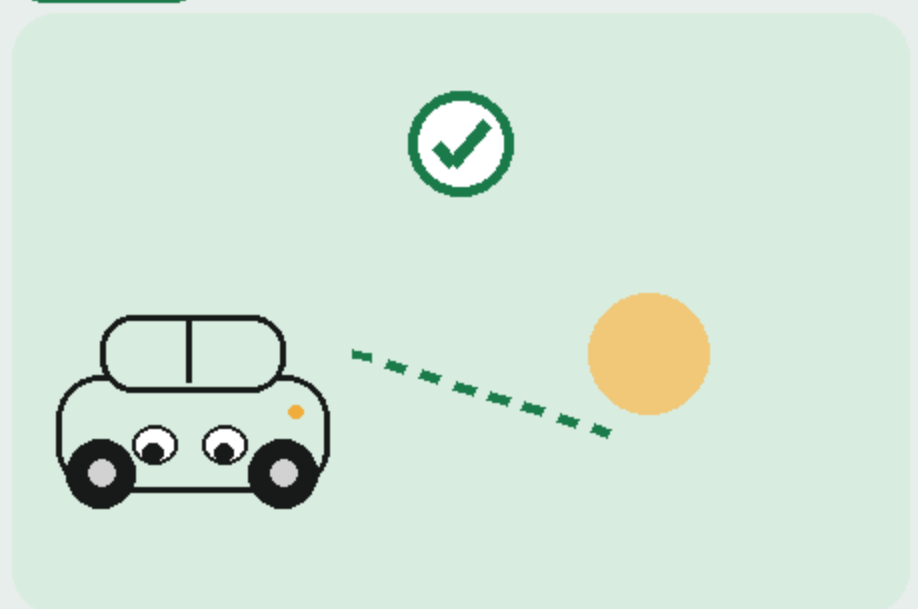
Instead of assuming traffic just freezes in place, this AI forecasts how the scene will evolve — and only swerves from its plan when a real, moving danger warrants it.

Like a driver who pictures where the other car is heading, not just where it is right now.

OLD



NEW

**Fewer needless swerves, safer real-hazard catches**

RiskWorld

03

EXIT 03 • EV BATTERY HEALTH

Reads battery health while it's still charging

Checking a battery's true health normally means pulling it aside for a slow, separate discharge test. This AI predicts the same health signal straight from the ordinary charging pattern.

Like a doctor reading your pulse during a routine checkup, instead of booking a separate test.

OLD



NEW

**Real-time health checks, zero downtime**

ChargeIC-LSTM

04

EXIT 04 • V2X MESSAGE RECOVERY

Guesses what a garbled safety message probably said

When a wireless safety broadcast arrives corrupted, this AI uses its own prediction of where the car probably is to help fill in the missing pieces and rescue the message.

Like finishing a friend's sentence when their voice cuts out mid-call.

OLD



NEW

**Rescues messages that would've been lost**

V2X-LDPC-Recovery

05

EXIT 05 • WORLD MODELS

Imagines what's coming, then hands it to the planner

One shared imagination of the near future feeds directly into the driving decision — flowing only one way, so daydreaming about the future never distorts what the car actually plans to do.

Like a co-pilot who narrates the road ahead, while the driver still makes every call.

OLD



NEW



One imagination, one-way into the plan

ForeDrive

06

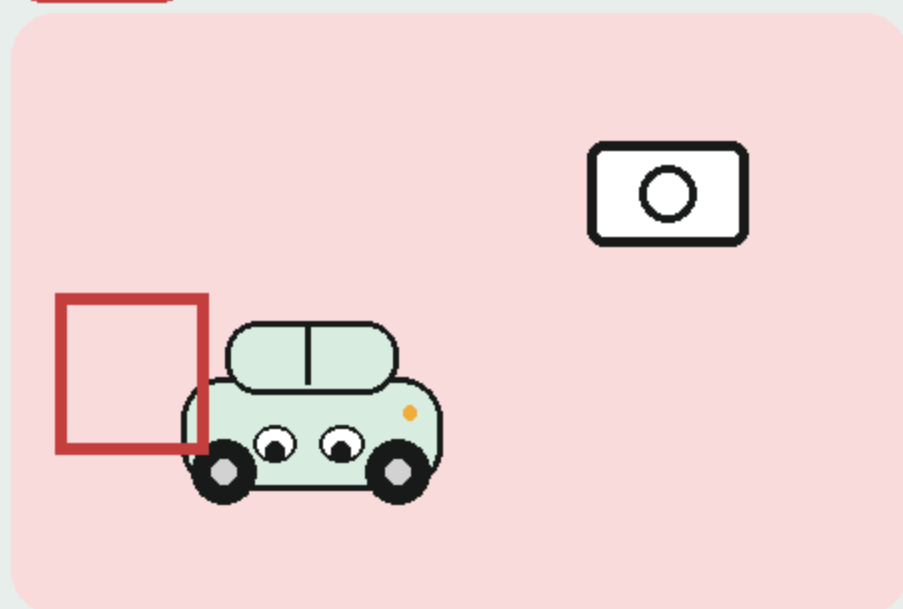
EXIT 06 • LATENCY COMPENSATION

Aims for where the world will be, not where it was

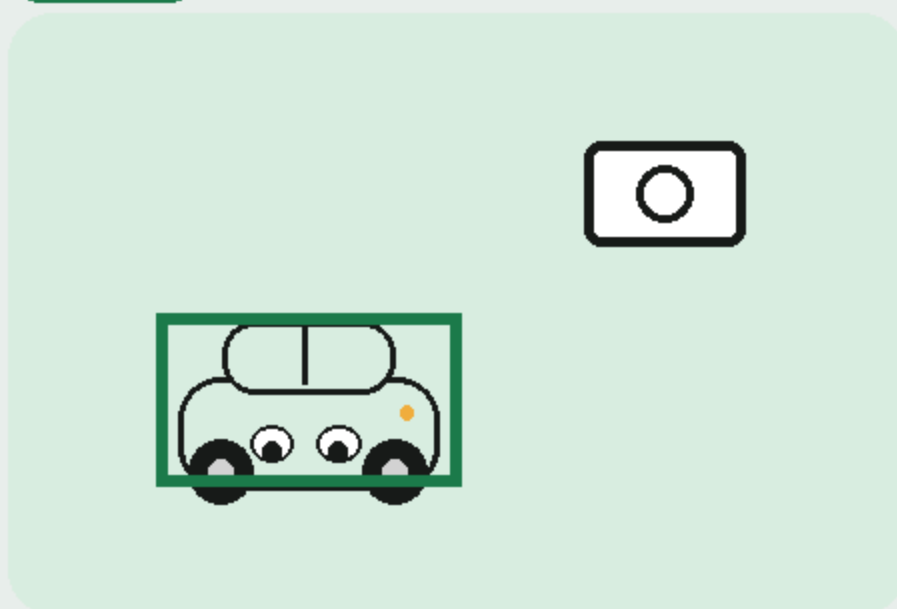
A detector takes real time to think, so by the time its answer arrives, the scene has moved on. This AI predicts forward to the exact moment its own answer will land.

Like leading a moving target when you throw, not aiming at where they used to stand.

OLD



NEW

**Recovers accuracy lost to processing lag**

ChronoFuse

07

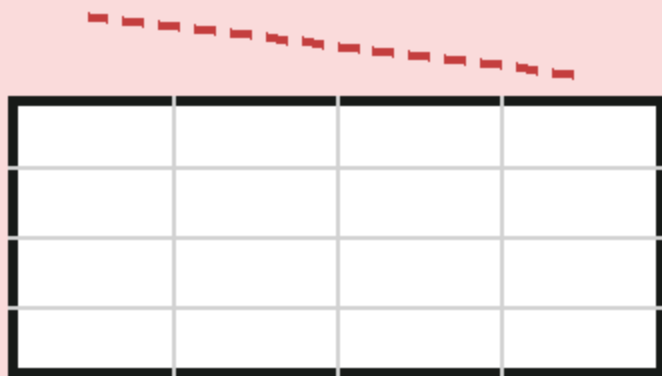
EXIT 07 · VISION-GROUNDED PLANNING

Double-checks its own plan against the cameras

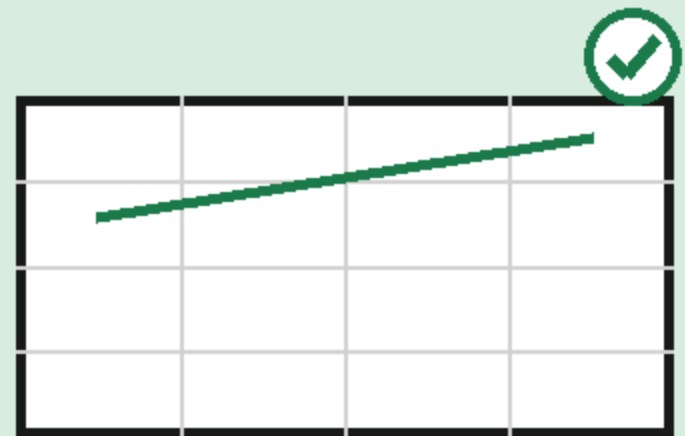
Rather than planning a path in the abstract, this AI projects each planned point back into every camera view and refines the route against what the cameras actually see.

Like glancing back at the road to confirm the route you just imagined.

OLD



NEW



A much tighter match to camera reality

S2Planner

08

EXIT 08 • MAP MEMORY

Remembers the road from the last time it drove here

Every drive doesn't have to start from zero. This AI recalls a map built from earlier trips down the same road — no LiDAR required — to fill in whatever the cameras miss today.

Like a commuter who already knows where the pothole is, even in the fog.

OLD



NEW



Map memory more than doubles road-layout accuracy

PriorMapBEVNet

Want the code?

Full write-ups, tested PyTorch implementations, and simulation GIFs for all 23 systems so far — free and open on GitHub.

`github.com/Lalithprabu/
automotive-ai-daily-reviews`

∞ **See the full repo**

